

Hamzah Marie

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EDUCATION

College of Staten Island

Bachelor of Science in Computer Science

Staten Island, NY

Expected Dec 2026

EXPERIENCE

New York City Office of Management and Budget (OMB)

Software Development Intern / College Aide

New York, NY

Jun 2026 – Present

- Engineered AIMS, a FastAPI and DuckDB-based asset management platform for 6,000+ NYC municipal assets, using a layered repository-service-router architecture for condition assessment and portfolio analysis.
- Developed a hierarchical scoring pipeline aggregating condition, coverage, criticality, and observation data across components and disciplines to compute facility scores and grades for 6,000+ assets.
- Built geospatial and analytics interfaces with Leaflet.js, HTMX, Jinja2, and JavaScript, including clustered asset mapping, BBL parcel boundaries, BIN building footprints, filtering, drilldowns, pagination, and CSV exports.
- Designed ETL and SQL workflows for multi-year reconciliation and asset-condition data using parameterized queries, window functions, deduplication, and bulk-processing scripts.

CUNY Tech Prep

Data Science Fellow

Hybrid

Jun 2025 – May 2026

- Selected for a competitive year-long Data Science fellowship serving students across 11 CUNY senior colleges.
- Built machine learning workflows in Python, Pandas, NumPy, and SQL covering data preparation, feature engineering, and model evaluation.

Students Who Code (Research Foundation)

Software Development Engineering Intern

Remote

Jun 2023 – Aug 2023

- Developed REST APIs with Node.js and MongoDB supporting data for 500+ students across 10+ courses.
- Built a quiz engine with score tracking and attempt history and validated 20+ API endpoints using Postman.

PROJECTS

Sports Management System | *FastAPI, PostgreSQL, AWS S3, HTMX, Jinja2*

Dec 2025 – Present

- Built a multi-tenant platform for sports organizations supporting participant enrollment, attendance, packages, waivers, and administrative workflows.
- Designed normalized PostgreSQL schemas and service-layer authorization for role-based access and organization-level data isolation.

Steam Engine - Game Recommendation Platform | *Next.js, React, FastAPI, Python*

Nov 2025 – Dec 2025

- Built a recommendation platform serving personalized results across a catalog of 82K+ Steam games.
- Implemented TF-IDF vectorization and cosine-similarity retrieval with FastAPI for low-latency content-based recommendations.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, TypeScript, HTML/CSS

Frameworks: FastAPI, HTMX, Jinja2, React, Next.js, Node.js, Express.js

Databases: PostgreSQL, SQL Server, DuckDB, MongoDB

Tools: Git, GitHub Actions, AWS S3, Linux, Bash, Postman, Pytest

Concepts: REST APIs, CI/CD, OAuth, JWT

LEADERSHIP & ACTIVITIES

Computer Science Club

Academic Tutor (Volunteer)

College of Staten Island

2024 – Present

- Mentor students in Data Structures, Algorithms, LeetCode, and technical interview preparation.